
Equivariant Compression of Quantum Operator Representations

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Abstract

1 Electronic structure is being increasingly prioritized in atomistic machine learning
2 (ML), as it offers a direct path to predicting diverse properties beyond single-
3 property ML surrogates. However, quantum mechanical (QM) data, such as matrix
4 representations of the electronic Hamiltonian, are inherently high-dimensional.
5 This makes standard operations, such as diagonalization, for obtaining eigenspectra
6 and other observables prohibitively expensive, and also makes the ML of such data
7 computationally demanding. In this work, we ask whether QM data can benefit
8 from classical compression strategies that are widely used in other fields. Taking
9 the example of effective single-particle Hamiltonians, we analyze how these tech-
10 niques interact with the symmetric matrix structure and affect derived observables.
11 We introduce an equivariant compression framework that preserves symmetry
12 while reducing dimensionality, allowing quantum observables computed from the
13 compressed representation to closely reproduce those from the full Hamiltonian.
14 Our results underscore the need for physics-aware compression techniques and
15 pave new directions for scalable, structure-preserving ML for electronic structure.

16 1 Introduction

17 Machine learning (ML) has transformed our approach to modeling complex physical systems,
18 paving the way for newer computational pipelines, such as hybrid ML–quantum workflows for
19 property prediction. Atomistic modeling, in particular, has progressed from learning interatomic
20 potentials [1–9] and surrogate models for individual properties [10, 11] to directly targeting electronic
21 structure itself. By targeting the fundamental ingredients of quantum mechanics, such as effective
22 single-particle Hamiltonians, ML can simultaneously optimize multiple observables within a single
23 framework, overcoming the limitations of separate property-specific surrogates.

24 Within these workflows, ML can take ingredients such as effective one-electron Hamiltonians as inputs
25 to predict a wide range of molecular properties [12–14] or entirely replace the traditional quantum
26 calculation [15–28]. The promise of unified modeling is, however, accompanied by the challenge
27 of representing and learning quantum mechanical entities, which are inherently higher-dimensional
28 than atomistic properties. For instance, matrix representations of the electronic Hamiltonian on
29 atomic basis sets scale quadratically with their size. This rapid growth makes standard quantum
30 mechanical operations computationally expensive and also poses a practical challenge for storing,
31 sharing, and training ML models on these datasets. Consequently, there is growing interest in
32 strategies to determine effective or downfolded representations of quantum mechanical operators
33 without sacrificing their essential physical structure.

34 One strategy is to explicitly construct a low-rank symmetry-adapted projected Hamiltonian
35 (SAPH)[20], which can then be used as a reduced target for ML. While this method accurately
36 preserves a subset of eigenvalues, it leads to unphysical predictions of other observables [29].

Refs. 30, 29 address this issue by exploiting differentiable pipelines to learn a compressed Hamiltonian by minimizing the error on several observables computed from it against their reference values from a larger basis set calculation.

In other domains routinely faced with high-dimensional data, classical compression techniques are reliable tools for handling data. For instance, image compression can reduce matrix dimensions by up to $25\times$ by exploiting problem-specific features [31, 32]. Techniques such as the Discrete Cosine Transform (DCT) enable efficient storage and reconstruction by using a fixed set of basis functions. While classical compression algorithms are typically optimized for storage rather than for dimensionality reduction, they have also been successfully applied in this context [31]. However, naive application of these methods on QM data can compromise physical symmetries, such as Hermiticity and rotational equivariance, leading to undesirable deviations in derived observables.

In this work, we analyze the interplay of several analytical compression techniques with the mathematically rich structure of QM data with the goal of designing compression strategies that lead to the most information-dense representations while preserving the underlying structure of the data. Although our focus here is on electronic Hamiltonians, the approach naturally extends to other QM operators and provides a strategy to balance cost–accuracy trade-offs in their modeling.

2 Background

Electronic Hamiltonian Operator. The electronic Hamiltonian operator is central to QM as it encodes all ground-state properties of a given atomic configuration. In practice, many quantum-mechanical theories and software packages work with a matrix representation of the operator in a finite basis of atom-centered orbitals (AOs), each of which is characterized by the chemical species a of the atom, radial (n) and angular quantum numbers (l, m). The matrix elements between $\mu \equiv (anlm)$ and $\nu \equiv (a'n'l'm')$ centered on atoms i and j in structure A , can then be represented as $\mathbf{H}_{Aij}^{anlm, a'n'l'm'} = \langle anlm | \hat{H} | a'n'l'm' \rangle$. One-electron eigenvalues ϵ (molecular orbital (MO) energies) and MO coefficients ($\mathbf{C}$) can be obtained by solving the generalized eigenvalue problem,

$$\mathbf{H}\mathbf{C} = \mathbf{S}\mathbf{C} \text{diag } \epsilon, \quad (1)$$

Ground state properties, such as the Löwdin atomic charges q_i , can then be computed by first computing the density matrix, $\rho = \mathbf{C} \text{diag}(\mathbf{f})\mathbf{C}^\dagger$, with occupation of each MO in $\mathbf{f}$, followed by

$$q_i = Z_i - \sum_{\mu \in i} \sum_{\nu \in i} (\mathbf{S}^{1/2} \rho \mathbf{S}^{1/2})_{\mu\nu}. \quad (2)$$

In addition to satisfying Hermiticity, $\mathbf{H} = \mathbf{H}^\dagger$, the matrix representation of the Hamiltonian also transforms under rotations as a consequence of the angular nature of AOs. For each pair of angular quantum numbers (l, l'), associated with the radial functions (n, n') and atoms i and j , $\mathbf{H}_{Aij}^{bmm'}$, the angular components are coupled using Clebsch–Gordan coefficients to obtain equivariant outputs indexed by L ,

$$\mathbf{H}_{Aij}^{b_{\text{symm}}, M} = \sum_{mm'} \langle lm; l'm' | LM \rangle \mathbf{H}_{Aij}^{b, mm'}, \quad (3)$$

where we use the shorthand $\mathbf{b} = (al, a'l')$ and $\mathbf{b}_{\text{symm}} = (al, a'l', sL)$ to denote the combined set of indices for the angular and radial basis functions, as well as the chemical species of the atoms. $\langle lm; l'm' | LM \rangle$ are the Clebsch–Gordan coefficients, s denotes inversion parity. The electronic Hamiltonian is, thus, a mathematically constrained object rather than an arbitrary dense matrix. In the following, we review a class of matrix compression techniques, and analyze to what extent they preserve the structure of the matrix and the physical properties derived from it.

Discrete Cosine Transform. Similar to the symmetry-adapted block decomposition in (3), popular data compression methods [33–35] also project matrices onto an orthogonal basis using transformations such as Discrete Cosine Transform (DCT) [36–38]. A two-dimensional DCT of an input matrix results in a matrix whose elements are coefficients of a data-independent basis of cosines of increasing frequency, such that coefficients in the top-left corner of the matrix correspond to low-frequency components, and coefficients in the

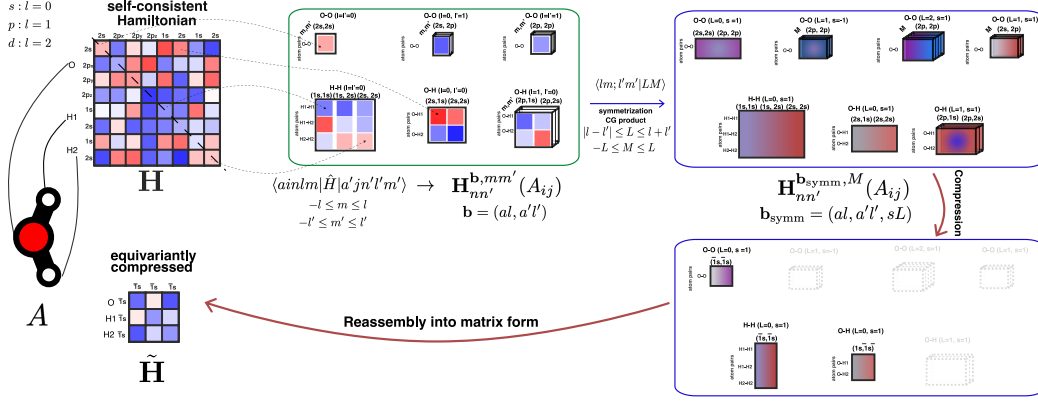


Figure 2: Schematic illustrating the decomposition of an input self-consistent Hamiltonian matrix into blocks $\mathbf{H}_{A_{ij}}^{b,mm'}$ labeled by angular quantum numbers [20, 39], followed by symmetrization via a Clebsch-Gordan tensor product. Compression to an effective representation $\tilde{\mathbf{H}}$ corresponds to retaining only the subset of matrix elements allowed by symmetry of the smaller effective basis.

bottom-right corner correspond to high-frequency components. This structure enables compression strategies such as high-frequency pruning (DCT-H), which removes the bottom-right frequency bands, and low-frequency pruning (DCT-L), which removes the top-left bands [31] (Figure 1). We collectively reference these methods as 2D compression. However, when applied to the Hamiltonian, these methods are agnostic to the orbital-resolved structure of the operator. Hence, they obscure the chemically or physically meaningful structure associated with the resulting matrix elements. The compressed matrices, however, can still be diagonalized and their eigenvalues and eigenvectors used to obtain approximate physical observables. In the following, we propose a framework to leverage these frequency-based compression techniques while retaining the symmetry-adapted decomposition of the Hamiltonian.

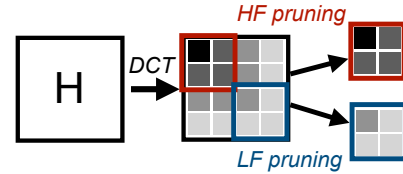


Figure 1: Discrete Cosine Transform (DCT) represents data in the frequency domain based on a range of low and high-frequency cosine functions. LF: Low-frequency. HF: High-frequency.

3 Methods

Since the physical symmetries of the Hamiltonian are naturally expressed through the symmetry-adapted decomposition (3), we compress each block independently along a single dimension, preserving both the atomic indexing and the $O(3)$ symmetry. Each block can then be compressed by a naive truncation, singular-value decomposition (SVD), or a one-dimensional version of the DCT method described above. We collectively reference these methods as 1D compression. For both 1D and 2D compression, we perform a single DCT over entire matrix/vector¹. As shown in Fig. 2, all of these approaches correspond to eliminating some blocks ($\mathbf{H}^{b_{\text{symm}}}$) or reducing the size of each block ($\mathbf{b}$), based on the size and symmetries of the target reduced effective basis. In this work, we focus on SVD, as well as, a one-dimensional DCT, noting that these methods are not specifically optimized for the downstream properties and may only partially capture global properties due to the independent treatment of each block. A possible extension is to apply similar compression strategies to functions of the matrix, such as algebraic powers, which we leave for future work. Another risk of treating blocks independently is that, even when the compressed basis matches the original dimension, each block can be offset by separate unitary transformations, yielding a non-unitary relationship between the original and reassembled matrices. However, this problem is resolved by fixing an overall gauge.

¹This is unlike how popular storage formats such as JPEG operate on images, which decomposes the image into smaller spatial patches, corresponding to adjacent submatrices and applies DCT to each patch separately.

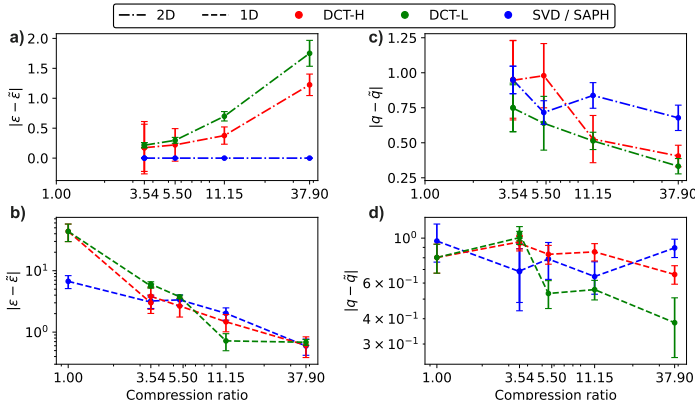


Figure 3: MAE on frontier MO eigenvalues and Löwdin charges obtained through compressed matrices obtained by 2D (matrix) (dash-dot lines) and 1D (block-wise) (dashed lines) compression. Compression ratios are defined as the average ratio of the matrix dimension in the TZVP basis to that in a smaller basis across all molecules in the dataset, corresponding to 1, 3.54, 5.50, 11.15, and 37.90 for TZVP, SVP, DZVP, 6-31G, and STO-3G, respectively. We also compute SAPH [40] which is an equivariant 2D compression.

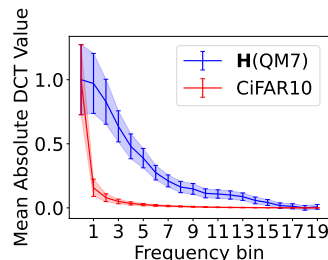


Figure 4: Normalized mean absolute value of frequency coefficients. Frequencies increase with the frequency index. For the **H** dataset, frequencies decay more slowly than for natural image data, motivating a re-examination of image-inspired compression and ML techniques and their adaptation modeling QM data.

112 4 Results and Discussion

113 We evaluate the performance of these compression techniques on a dataset of 1000 molecules,
 114 containing C, H, N, and O atoms, from the QM7b dataset [41], comprising Fock and overlap matrices
 115 computed at the B3LYP/def2-TZVP level in PySCF [42]. We benchmark the compression across a
 116 range of resolutions corresponding to smaller basis sets, namely, def2SVP, cc-PVDZ, DZVP, 6-31G,
 117 and STO-3G, noting that the compressed matrices resulting from 2D DCT may be comparable in size
 118 but cannot be truly interpreted in terms of specific orbital labels. For all experiments, we report the
 119 mean absolute error (MAE) on a set of frontier MO eigenvalues and Löwdin atomic charges computed
 120 from the compressed matrices against the reference calculations (def2-TZVP basis). In computing
 121 (2), we use the exact overlaps computed on the corresponding reduced basis, since compressing the
 122 def2-TZVP overlap matrix directly can easily compromise its positive definiteness.

123 Fig. 3a) aligns most closely with intuition, DCT-H which preserves low frequencies, better approx-
 124 imates global functions of the matrix (eigenvalues) than DCT-L, which prunes low frequencies.
 125 SAPH corresponds to zero errors as it preserves a subset of eigenvalues by design. In comparison,
 126 3b) suffers from the gauge fixing problem and shows a systematic improvement on frontier MO
 127 eigenvalues with compression across all methods. However, 1D DCTs are more accurate than their
 128 corresponding 2D counterparts at the highest compression ratio. Frequency based compressed matri-
 129 ces approximate charges much better than SAPH, especially at the highest compression ratio using
 130 both 2D and 1D equivariant frequency compression (3c, d)). The decreasing trend of the MAE with
 131 basis set size warrants further investigation, especially as the compressed Hamiltonian interacts with
 132 the exact overlap matrix in the computation of properties, causing the effects of compression and
 133 basis truncation to be intrinsically intertwined. We are exploring analogous compression strategies
 134 for orthogonalized Hamiltonians.

135 Finally, we use the 2D frequency-based compression to analyze the structure of the Hamiltonian
 136 matrix, comparing it against image data using the popular CiFAR10 dataset [43] in Figure 4. As
 137 expected, we notice a clear difference between the frequency distribution of QM data, which must be
 138 accounted for, especially in light of recent interest in machine learning electronic structure through
 139 tools that are more naturally applied to image data [28, 44–47]. While we demonstrated a strategy for
 140 compressing electronic Hamiltonians that preserves symmetry, this work highlights the opportunity
 141 to link molecular properties and eigenvalues with matrix frequencies and symmetries, guiding the
 142 selection of Hamiltonian functions to compress and combining classical and ML techniques for
 143 interpretable, efficient representations of quantum data.

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